

The cross-rank commutator $[R_e(t), R_f(t)]$ in closed form

Clio

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Abstract

Let $R_e(t)$ be the operator on the ring of symmetric functions that adds a connected e -ribbon with weight t^{ht} . We compute the commutator $[R_e(t), R_f(t)]$ for all $e, f \geq 1$ in closed form. It has exactly two pieces, separated by how many beads of the Maya diagram they transport:

$$[R_e(t), R_f(t)] = -\frac{1+t}{t} \left(\Phi_{e,f} + (t-1) \Psi_{e,f} \right),$$

where $\Phi_{e,f}$ is a one-bead operator — a dressed $(e+f)$ -ribbon move — and $\Psi_{e,f}$ is a genuine two-bead operator. Both are written explicitly as undeformed Clifford expressions.

Four consequences. (i) The divisibility of $[R_e, R_f]$ by $(1+t)$, previously proved by evaluating at the anchor $t = -1$, becomes *structural*: every matrix element is a difference of two routes whose ht-statistics differ by exactly 1, so the element is a multiple of $t^{N-1}(1+t)$ before any cancellation. (ii) $\Psi_{e,f} = 0$ if and only if $\min(e, f) = 1$; consequently $[R_1(t), R_f(t)]$ — Zabrocki's add-one-cell operator — is purely one-bead and we give it in closed form. (iii) The one-bead coefficient has a clean combinatorial reading: it compares whether the ribbon *turns* at position e against whether it turns at position f . (iv) The conjecture the session set out to test is **false**: $\partial_t[R_e, R_f]|_{t=-1}$ is not a multiplication operator, so it is not a Lie bracket on the span of the power sums; the family $\{R_e(t)\}$ spans no Lie algebra, and the Lie algebra it generates contains operators of bead-number 3.

Two gaps in the first draft are closed in §7: $\Psi_{e,f} \neq 0$ for *all* $e, f \geq 2$, by a uniform witness; and the three-bead sector of $[R_e, [R_f, R_g]]$ factorises as $t^P(t^{\kappa_{32}} - t^{-\kappa_{32}})(t^{\kappa_{21} + \kappa_{31}} - t^{-(\kappa_{21} + \kappa_{31})})$, which makes the bead-3 statement uniform. What remains open is stated in §8; the honest headline there is that this session had no network, so the $W_{1+\infty}$ / Bloch–Okounkov novelty question is *unchecked* and nothing here claims priority.

Verified by three code-disjoint engines: 1005/1005 (combinatorial vs. Clifford, $|\lambda| \leq 8$, $1 \leq e < f \leq 6$) and 750/750 (combinatorial vs. route-counting, including $e = f$).

1 Conventions and what is assumed

We keep the conventions of [1] verbatim.

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$ with Schur basis $\{s_\lambda\}$ and Hall inner product. For a partition λ the *Maya set* is $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$; its elements are *beads*, its complement *holes*, and $m(x) = [x \in M]$. We write $|\lambda\rangle$ for s_λ and identify Λ with the charge-0 semi-infinite wedge $\mathcal{F}$ spanned by the $|M\rangle$, with undeformed free fermions $\{\psi_a, \psi_b^*\} = \delta_{ab}$ and $n_a = \psi_a \psi_a^*$. For $e \geq 1$,

$$R_e(t) s_\lambda = \sum_{\mu} t^{\text{ht}(\mu/\lambda)} s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected border strip of size e and $\text{ht} = (\#\text{rows}) - 1$. We abbreviate $N_{(x,y)} = \sum_{x < j < y} n_j$ and $\#(M \cap (x, y))$ for its eigenvalue.

We use one input, proved in [1, Lem. 2.3, Lem. 2.4, Thm. 3.3]:

Proposition 2 (abacus form of R_e ; [1]). *Call the pair (b, g) a legal g -move on M if $b \in M$ and $b + g \notin M$, and write $M \cdot (b, g) := (M \setminus \{b\}) \cup \{b + g\}$. Then*

$$R_g(t) |M\rangle = \sum_{(b,g) \text{ legal}} t^{\#(M \cap (b, b+g))} |M \cdot (b, g)\rangle,$$

a finite sum; the map $b \mapsto M \cdot (b, g)$ is a bijection from legal g -moves onto the μ with μ/λ a connected g -ribbon, and $\text{ht}(\mu/\lambda) = \#(M \cap (b, b + g))$. Equivalently, as operators, $R_g(t) = \sum_{b \in \mathbb{Z}} \psi_{b+g} \psi_b^ (-t)^{N(b, b+g)}$.*

Convention 3 (bead-number grading). An operator X on $\mathcal{F}$ has *bead-number* r if $\langle M' | X | M \rangle = 0$ whenever $|M \triangle M'| \neq 2r$. Since $\{|M\rangle\}$ is a basis, every operator with locally finite matrix decomposes uniquely as $\sum_{r \geq 0} X^{(r)}$ with $X^{(r)}$ of bead-number r . By Proposition 2 each R_g has bead-number 1, so a product $R_e R_f$ has bead-number parts only in degrees $r \leq 2$.

2 The matrix elements

Everything follows from an exhaustive count of the ways a product of an e -move and an f -move can reach a given target. This section uses no fermions.

Lemma 4 (the two interference indices). *Fix $e, f \geq 1$ and $b, c \in \mathbb{Z}$, and put $x = c - b$. Set*

$$\alpha = [c \in (b, b + e)], \quad \beta = [c + f \in (b, b + e)], \quad \alpha' = [b \in (c, c + f)], \quad \beta' = [b + e \in (c, c + f)],$$

and $k_{b,c}^{e,f} := \beta' - \alpha'$. Then

$$(\beta - \alpha) + (\beta' - \alpha') = [x = 0] - [x = e - f].$$

In particular $\beta - \alpha = -k_{b,c}^{e,f}$ unless $b = c$ or $b + e = c + f$.

Proof. In the variable x the four conditions read $\alpha = [0 < x < e]$, $\beta = [-f < x < e - f]$, $\alpha' = [-f < x < 0]$, $\beta' = [e - f < x < e]$. The intervals $(0, e)$ and $(-f, 0)$ together with the point 0 partition the integers of $(-f, e)$; so do $(-f, e - f)$ and $(e - f, e)$ together with the point $e - f$. Hence $\alpha + \alpha' + [x = 0] = [-f < x < e] = \beta + \beta' + [x = e - f]$, which is the claim. $\square$

The number $k_{b,c}^{e,f}$ has a direct meaning: it is the change in $\#(M \cap (c, c + f))$ caused by performing the e -move at $b \rightarrow +1$ if the moving bead lands inside the interval $(c, c + f)$, -1 if it leaves it, 0 otherwise.

Theorem 5 (matrix elements of the cross-rank commutator). *Let $e, f \geq 1$, let M be a Maya set, and let $M' \neq M$. Then $\langle M' | [R_e, R_f] | M \rangle = 0$ unless $|M \triangle M'| \in \{2, 4\}$, and:*

(1) One-bead sector. *If $M' = M \cdot (b, e + f)$ is a legal $(e + f)$ -move, write $N = \#(M \cap (b, b + e + f))$. Then*

$$\langle M' | [R_e, R_f] | M \rangle = (m(b + e) - m(b + f)) (1 + t) t^{N-1}.$$

(The exponent $N - 1$ is never negative on the support: if $N = 0$ then $m(b + e) = m(b + f) = 0$ and the coefficient vanishes.)

(2) Two-bead sector. *Suppose $|M \triangle M'| = 4$. If $e = f$ then $\langle M' | [R_e, R_f] | M \rangle = 0$. If $e \neq f$, then the element vanishes unless $M' = (M \setminus \{b, c\}) \cup \{b + e, c + f\}$ for a (then unique) pair*

b, c with $b, c \in M$, $b + e, c + f \notin M$ and $b, c, b + e, c + f$ pairwise distinct. In that case, with $P = \#(M \cap (b, b + e))$, $Q = \#(M \cap (c, c + f))$ and $k = k_{b,c}^{e,f}$,

$$\langle M' | [R_e, R_f] | M \rangle = t^{P+Q} (t^{-k} - t^k).$$

(Again no negative exponent occurs: $k = 1$ forces $P \geq 1$ and $k = -1$ forces $Q \geq 1$.)

Proof. That only $|M \triangle M'| \in \{0, 2, 4\}$ occurs is Convention 3; the case $M' = M$ is excluded because a route adds $e + f > 0$ cells, so $|\mu'| = |\mu| + e + f$.

By Proposition 2,

$$\langle M' | R_f R_e | M \rangle = \sum_{\text{routes}} t^{w_1 + w_2}, \quad (1)$$

summed over pairs of legal moves $M \xrightarrow{(x,e)} M_1 \xrightarrow{(y,f)} M'$, with $w_1 = \#(M \cap (x, x + e))$ and $w_2 = \#(M_1 \cap (y, y + f))$. (Note $R_f R_e$ applies R_e first.) We enumerate the routes.

Step 1: which routes cancel. We have $M' = (M \setminus \{x\} \cup \{x + e\}) \setminus \{y\} \cup \{y + f\}$. The four sites x, y (removed) and $x + e, y + f$ (added) leave $|M \triangle M'| = 4$ unless one removed site coincides with one added site. Since $x \neq x + e$ and $y \neq y + f$, the only possibilities are $y = x + e$ or $x = y + f$.

Step 2: the one-bead sector. Let $M' = M \cdot (b, e + f)$, so $b \in M$, $b + e + f \notin M$.

Route B ($y = x + e$). Then $M' = M \setminus \{x\} \cup \{x + e + f\}$, forcing $x = b$. Legality: $b \in M$ and $b + e \notin M$ for the first move; for the second, $y = b + e \in M_1$ holds automatically and $y + f = b + e + f \notin M_1$ is equivalent to $b + e + f \notin M$, which is given. So route B exists iff $m(b + e) = 0$. Its weight: $w_1 = \#(M \cap (b, b + e))$, and since the two sites at which M_1 differs from M , namely b and $b + e$, both lie outside the open interval $(b + e, b + e + f)$, we get $w_2 = \#(M \cap (b + e, b + e + f))$. Hence $w_1 + w_2 = \#(M \cap (b, b + e + f)) - m(b + e) = N$, using $m(b + e) = 0$.

Route C ($x = y + f$). Then $M' = M \setminus \{y\} \cup \{y + e + f\}$, forcing $y = b$ and $x = b + f$. Legality: $x = b + f \in M$ is required; $x + e = b + e + f \notin M$ is given; $y = b \in M_1$ holds since $b \in M$ and $b \neq b + f$; and $y + f = b + f \notin M_1$ holds because M_1 removed it. So route C exists iff $m(b + f) = 1$. Its weight: $w_1 = \#(M \cap (b + f, b + e + f))$, and since M_1 differs from M only at $b + f$ and $b + e + f$, both outside the open interval $(b, b + f)$, we get $w_2 = \#(M \cap (b, b + f))$. Hence $w_1 + w_2 = \#(M \cap (b, b + e + f)) - m(b + f) = N - 1$, using $m(b + f) = 1$.

Routes B and C are mutually exclusive as *route types* only in the sense that $y = x + e$ and $x = y + f$ cannot hold simultaneously ($e + f = 0$); both may nevertheless occur, for different x . Summing (1),

$$\langle M' | R_f R_e | M \rangle = (1 - m(b + e)) t^N + m(b + f) t^{N-1}.$$

Exchanging the roles of e and f (the target $M \cdot (b, e + f)$ is unchanged) gives

$$\langle M' | R_e R_f | M \rangle = (1 - m(b + f)) t^N + m(b + e) t^{N-1},$$

and subtracting,

$$\langle M' | [R_e, R_f] | M \rangle = (m(b + e) - m(b + f)) (t^N + t^{N-1}),$$

which is (1). For the parenthetical: $b + e$ and $b + f$ both lie in the open interval $(b, b + e + f)$ because $e, f \geq 1$; so $N = 0$ forces $m(b + e) = m(b + f) = 0$.

Step 3: the two-bead sector. Here no cancellation occurs, so by Step 1 the removed sites are $\{x, y\} = M \setminus M'$ and the added sites are $\{x + e, y + f\} = M' \setminus M$, all four distinct. Write $M \setminus M' = \{b, c\}$. If the e -move were made from c , then $c + e \in \{b + e, c + f\}$ would force $c = b$ or

$e = f$. So for $e \neq f$ the assignment $x = b, y = c$ is forced once we name b as the bead displaced by e ; the pair (b, c) is determined by M, M' because b is the unique element of $M \setminus M'$ with $b + e \in M' \setminus M$ (the alternative $b \mapsto c + f, c \mapsto b + e$ would give $c - b = f - e$ and $c - b = e - f$ simultaneously, hence $e = f$).

Legality: $b \in M, b + e \notin M$; then $c \in M_1$ iff $c \in M$ (as $c \neq b, c \neq b + e$), and $c + f \notin M_1$ iff $c + f \notin M$ (as $c + f \neq b, c + f \neq b + e$). Weights: $w_1 = P$, and

$$w_2 = \#(M_1 \cap (c, c + f)) = Q - [b \in (c, c + f)] + [b + e \in (c, c + f)] = Q + k.$$

So $\langle M' | R_f R_e | M \rangle = t^{P+Q+k}$. Running the same argument with e and f exchanged — now the f -move at c comes first — gives weights Q and $P + (\beta - \alpha)$, and $\beta - \alpha = -k$ by Lemma 4, since $b \neq c$ and $b + e \neq c + f$. So $\langle M' | R_e R_f | M \rangle = t^{P+Q-k}$ and the difference is as claimed.

If $e = f$ the two assignments $(x, y) = (b, c)$ and $(x, y) = (c, b)$ are both available in each order, and the two orders contribute the same multiset of weights, so the commutator vanishes; this also follows from $[R_e, R_e] = 0$.

Finally the sign conditions on exponents. If $k = 1$ then $b + e \in (c, c + f)$ and $b \notin (c, c + f)$; from $b < b + e < c + f$ and $b \notin (c, c + f)$ we get $b \leq c$, hence $b < c < b + e$, so $c \in M \cap (b, b + e)$ and $P \geq 1$. If $k = -1$ then $b \in (c, c + f)$, so $b \in M \cap (c, c + f)$ and $Q \geq 1$. $\square$

3 The closed operator form

Theorem 6 (cross-rank commutator). *For all $e, f \geq 1$, as operators on $\mathcal{F} \cong \Lambda$,*

$$[R_e(t), R_f(t)] = -\frac{1+t}{t} \left(\Phi_{e,f} + (t-1) \Psi_{e,f} \right)$$

where

$$\Phi_{e,f} = \sum_{b \in \mathbb{Z}} \psi_{b+e+f} \psi_b^* (-t)^{N(b, b+e+f)} (n_{b+f} - n_{b+e})$$

is a bead-number-1 operator (a dressed $(e+f)$ -ribbon move), and

$$\Psi_{e,f} = \sum_{b, c \in \mathbb{Z}} k_{b,c}^{e,f} \psi_{b+e} \psi_{c+f} \psi_b^* \psi_c^* (-t)^{N(b, b+e) + N(c, c+f)}$$

is a bead-number-2 operator, with $k_{b,c}^{e,f}$ as in Lemma 4. Both sums are locally finite.

Proof. Write $A_b = \psi_{b+e} \psi_b^*$, $C_c = \psi_{c+f} \psi_c^*$, $D_b = (-t)^{N(b, b+e)}$, $G_c = (-t)^{N(c, c+f)}$, so that $R_e = \sum_b A_b D_b$ and $R_f = \sum_c C_c G_c$ by Proposition 2.

Step 1: moving the dressings across. C_c transports a bead from c to $c + f$, so on its support the eigenvalue of $N(b, b+e)$ changes by $\beta - \alpha$ in the notation of Lemma 4; off its support both sides below vanish. Hence, as operators,

$$D_b C_c = (-t)^{\beta - \alpha} C_c D_b, \quad \text{and symmetrically} \quad G_c A_b = (-t)^{\beta' - \alpha'} A_b G_c.$$

Since D_b and G_c are both diagonal they commute, so

$$A_b D_b C_c G_c = u_{bc} A_b C_c D_b G_c, \quad C_c G_c A_b D_b = v_{bc} C_c A_b D_b G_c,$$

with $u_{bc} = (-t)^{\beta - \alpha}$ and $v_{bc} = (-t)^{\beta' - \alpha'} = (-t)^{k_{b,c}^{e,f}}$.

Step 2: the Clifford products. Using $\psi_j^* \psi_k = \delta_{jk} - \psi_k \psi_j^*$,

$$A_b C_c = \delta_{b,c+f} \psi_{b+e} \psi_c^* - \Theta_{bc}, \quad C_c A_b = \delta_{c,b+e} \psi_{c+f} \psi_b^* - \Theta_{bc},$$

where $\Theta_{bc} = \psi_{b+e} \psi_{c+f} \psi_b^* \psi_c^*$ is the *same* operator in both lines, because $\psi_{c+f} \psi_{b+e} \psi_c^* \psi_b^* = (-\psi_{b+e} \psi_{c+f})(-\psi_b^* \psi_c^*) = \Theta_{bc}$. Therefore

$$[R_e, R_f] = \underbrace{\sum_{b,c} (u_{bc} \delta_{b,c+f} \psi_{b+e} \psi_c^* - v_{bc} \delta_{c,b+e} \psi_{c+f} \psi_b^*) D_b G_c}_{\text{resonant}} - \underbrace{\sum_{b,c} (u_{bc} - v_{bc}) \Theta_{bc} D_b G_c}_{\text{quartic}}. \quad (2)$$

Step 3: the resonant part is $-\frac{1+t}{t} \Phi_{e,f}$. In the first sum $b = c + f$, i.e. $x = c - b = -f$; then $\alpha = [0 < -f < e] = 0$ and $\beta = [-f < -f < e - f] = 0$, so $u_{bc} = 1$. In the second sum $c = b + e$, i.e. $x = e$; then $\alpha' = [-f < e < 0] = 0$ and $\beta' = [e - f < e < e] = 0$, so $v_{bc} = 1$. Reindexing both by the moving bead b , and using

$$N_{(b,b+e)} + N_{(b+e,b+e+f)} = N_{(b,b+e+f)} - n_{b+e}, \quad N_{(b+f,b+e+f)} + N_{(b,b+f)} = N_{(b,b+e+f)} - n_{b+f},$$

the resonant part equals

$$\sum_b \psi_{b+e+f} \psi_b^* (-t)^{N_{(b,b+e+f)}} \left[(-t)^{-n_{b+f}} - (-t)^{-n_{b+e}} \right].$$

(The diagonal operators n_{b+e}, n_{b+f} commute with $\psi_{b+e+f} \psi_b^*$ because $b+e, b+f \notin \{b, b+e+f\}$, as $e, f \geq 1$.) Since $n^2 = n$ we have $(-t)^{-n} = 1 + ((-t)^{-1} - 1)n$, so the bracket equals $-\frac{1+t}{t} (n_{b+f} - n_{b+e})$, giving $-\frac{1+t}{t} \Phi_{e,f}$.

Step 4: the quartic part is $-\frac{(t-1)(1+t)}{t} \Psi_{e,f}$. Note first that $\Theta_{bc} = 0$ when $b = c$ (then $\psi_b^* \psi_c^* = 0$) and when $b + e = c + f$ (then $\psi_{b+e} \psi_{c+f} = 0$). Away from these two values of x , Lemma 4 gives $\beta - \alpha = -k_{b,c}^{e,f}$, hence

$$u_{bc} = (-t)^{-k}, \quad v_{bc} = (-t)^k, \quad u_{bc} - v_{bc} = (-t)^{-k} - (-t)^k = k \frac{t^2 - 1}{t}$$

for $k = k_{b,c}^{e,f} \in \{-1, 0, 1\}$ (check the three values directly; note $k \in \{-1, 0, 1\}$ because $\alpha', \beta' \in \{0, 1\}$).

Thus the quartic part of (2) equals $-\frac{t^2-1}{t} \Psi_{e,f} = -\frac{(t-1)(1+t)}{t} \Psi_{e,f}$.

Step 5: bead numbers. $\Phi_{e,f}$ visibly has bead-number 1. For $\Psi_{e,f}$: whenever $k_{b,c}^{e,f} \neq 0$ and $b \neq c$, the four sites $b, c, b+e, c+f$ are pairwise distinct. Indeed $b \neq b+e$ and $c \neq c+f$; if $c = b+e$ then $x = e$ and we computed $\alpha' = \beta' = 0$, so $k = 0$; if $b = c+f$ then $x = -f$ and again $\alpha' = \beta' = 0$, so $k = 0$; and if $b+e = c+f$ then $x = e-f$, where $\alpha' = [-f < e-f < 0] = 0$ and $\beta' = [e-f < e-f < e] = 0$, so $k = 0$. The remaining degenerate case $b = c$ has $\Theta_{bc} = 0$. Hence every surviving summand transports two beads.

Local finiteness of both sums is inherited from Proposition 2. Finally, the matrix elements of the right-hand side agree with Theorem 5: this is the consistency check recorded in §6, and it is what pins the fermionic signs. $\square$

Corollary 7 (structural divisibility). *$(1+t)$ divides every matrix element of $[R_e(t), R_f(t)]$, in $\mathbb{Z}[t]$. More precisely, in the one-bead sector every element is $\pm(1+t)t^{N-1}$ or 0, and in the two-bead sector $\pm(1-t^2)t^{P+Q-1}$ or 0.*

Proof. Immediate from Theorem 5, since $t^{-k} - t^k = \mp(t^2 - 1)t^{-1}$ for $k = \pm 1$. $\square$

Remark 8 (why the $(1+t)$ is there). In [1, Cor. 5.6] this divisibility was proved by evaluating at the anchor: $R_e(-1) = M_{pe}$, multiplications commute, so every coefficient vanishes at $t = -1$. That argument is correct but tells one nothing about the quotient. Theorem 5 replaces it by a mechanism. In the one-bead sector there are exactly two routes from λ to μ , the “in-order push” $b \rightarrow b+e \rightarrow b+e+f$ and the “leapfrog” $b+f \rightarrow b+e+f$ followed by $b \rightarrow b+f$; their ht-statistics are N and $N-1$, differing by exactly 1; and exchanging e and f exchanges *which* route is available while preserving the pair of exponents $\{N-1, N\}$. The commutator is therefore forced to be a multiple of $t^{N-1} + t^N$. The same holds in the two-bead sector with the pair $\{P+Q-1, P+Q+1\}$, whence the factor $t^{-k} - t^k$. The divisibility is not a coincidence at a special point; it is the statement that reordering the two ribbons costs exactly one unit of height.

4 The case $e = 1$: Zabrocki’s add-one-cell operator

Corollary 9. $\Psi_{e,f} = 0$ if and only if $\min(e, f) = 1$. Consequently, for every $f \geq 1$, $[R_1(t), R_f(t)]$ has bead-number 1 and

$$[R_1(t), R_f(t)] = -\frac{1+t}{t} \sum_{b \in \mathbb{Z}} \psi_{b+f+1} \psi_b^* (-t)^{N(b, b+f+1)} (n_{b+f} - n_{b+1}),$$

equivalently, on partitions,

$$[R_1(t), R_f(t)] s_\lambda = (1+t) \sum_{\mu} (m(b+1) - m(b+f)) t^{\text{ht}(\mu/\lambda)-1} s_\mu,$$

the sum over μ with μ/λ a connected $(f+1)$ -ribbon, b its bead move.

Proof. Take $e = 1$. A summand of $\Psi_{1,f}$ survives only if $k_{b,c}^{1,f} \neq 0$ and $\Theta_{bc} \neq 0$. Now $k = \beta' - \alpha'$ with $\alpha' = [-f < x < 0]$ and $\beta' = [1-f < x < 1]$, where $x = c - b$. So $k = +1$ forces $x = 0$ (the only integer in $(1-f, 1)$ not in $(-f, 0)$), where $\Theta_{bc} = 0$; and $k = -1$ forces $x = 1-f$ (the only integer in $(-f, 0)$ not in $(1-f, 1)$), i.e. $b+e = b+1 = c+f$, where again $\Theta_{bc} = 0$. Hence $\Psi_{1,f} = 0$, and by antisymmetry $\Psi_{f,1} = 0$.

Conversely, if $e, f \geq 2$, take c arbitrary and $b = c + f - 1$. Then $x = 1 - f \leq -1 < 0$ and $x > -f$, so $\alpha' = 1$; and $b+e = c+f+e-1 > c+f$ since $e \geq 2$, so $\beta' = 0$ and $k = -1$. The four sites are distinct: $b \neq c$ (as $f \geq 2$), $b+e \neq c+f$ (as $e \geq 2$), $c \neq b+e$, $b \neq c+f$. So $\Psi_{e,f}$ has a nonzero summand; that the operator itself is nonzero is Corollary 12 below, proved from Theorem 5.

The displayed formulas are Theorem 6 with $\Psi = 0$, and Theorem 5(1) with $e = 1$ combined with Proposition 2. $\square$

The one-bead coefficient has a purely combinatorial reading. Recall (Proposition 2) that a connected g -ribbon μ/λ corresponds to a bead move $b \rightarrow b+g$; read its g cells from the southwest tail to the northeast head and let $\tau_j(\mu/\lambda) \in \{0, 1\}$ record whether the j -th and $(j+1)$ -st cells lie in *different* rows, for $1 \leq j \leq g-1$.

Lemma 10. $\tau_j(\mu/\lambda) = m(b+j)$ for $1 \leq j \leq g-1$.

Proof. This is the cell-by-cell refinement of [1, Lem. 2.4]. In that proof the rows of μ that grow are exactly those attached to the beads of M in the open interval $(b, b+e)$, each by one cell, together with the moved bead’s own row; reading the ribbon from its tail, the j -th internal edge is a row change precisely when the site $b+j$ carries such a bead. (Verified independently on 1178 ribbons, $|\lambda| \leq 7$, $g \leq 6$: see §6.) $\square$

Corollary 11 (the commutator compares two turning positions). *Let μ/λ be a connected $(e+f)$ -ribbon. Then*

$$\langle s_\mu, [R_e(t), R_f(t)] s_\lambda \rangle = (1+t) t^{\text{ht}(\mu/\lambda)-1} (\tau_e(\mu/\lambda) - \tau_f(\mu/\lambda)).$$

That is: the one-bead part of $[R_e, R_f]$ sees only whether the ribbon turns after its e -th cell against whether it turns after its f -th cell. It vanishes on every ribbon that behaves the same way at both positions.

Proof. Theorem 5(1) and Lemma 10. □

5 What the family is not

Corollary 12 (the family spans no Lie algebra). *For $e, f \geq 2$ the two-bead part of $[R_e(t), R_f(t)]$ is nonzero; hence $\Psi_{e,f} \neq 0$, and $[R_e, R_f]$ is not a $\mathbb{Q}(t)$ -linear combination of bead-number-1 operators. For $e = 1 < f$ the commutator has bead-number 1 but is still not in $\text{span}_{\mathbb{Q}(t)}\{R_g(t)\}$. In all cases with $e \neq f$,*

$$[R_e(t), R_f(t)] \notin \text{span}_{\mathbb{Q}(t)}\{R_g(t) : g \geq 1\}.$$

Proof. For the first claim it suffices to exhibit one nonzero two-bead matrix element. Take $\lambda = \emptyset$, so $M = \mathbb{Z}_{<0}$, and $e = 2, f = 3, b = -1, c = -3$. Then $b, c \in M$; $b + e = 1 \notin M$; $c + f = 0 \notin M$; and the four sites $-1, -3, 1, 0$ are pairwise distinct, so this is a legitimate two-bead target. Here $x = c - b = -2$, $\alpha' = [-3 < -2 < 0] = 1$, $\beta' = [-1 < -2 < 2] = 0$, so $k = -1$. Also $P = \#(M \cap (-1, 1)) = \#\{0\} \cap M = 0$ and $Q = \#(M \cap (-3, 0)) = \#\{-2, -1\} = 2$. Theorem 5(2) gives the coefficient $t^2(t^1 - t^{-1}) = t^3 - t$, on the target $M' = M \setminus \{-1, -3\} \cup \{1, 0\}$, i.e. $\mu = (2, 2, 1)$. Indeed $[R_2, R_3]s_\emptyset$ contains $(t^3 - t)s_{(2,2,1)}$, and $(2, 2, 1)/\emptyset$ requires two beads to move. A bead-number-1 operator has no such matrix element.

For $e = 1 < f$: $[R_1, R_f]$ raises the degree $|\lambda|$ by exactly $f + 1$, whereas R_g raises it by g ; so in any expression $[R_1, R_f] = \sum_g c_g R_g$ only $g = f + 1$ can have $c_g \neq 0$, and $[R_1, R_f]$ would be a scalar multiple of R_{f+1} . But by Corollary 9 its matrix element at a ribbon vanishes whenever $\tau_1 = \tau_f$, while the corresponding matrix element of R_{f+1} is $t^{\text{ht}} \neq 0$. Explicitly for $f = 2$: $[R_1, R_2]s_\emptyset = (1+t)s_{(2,1)}$, whereas $R_3s_\emptyset = s_{(3)} + t s_{(2,1)} + t^2 s_{(1,1,1)}$; the supports differ. The last display combines the two cases (for $e, f \geq 2$ the span consists of bead-number-1 operators, which the commutator is not). □

Corollary 13 (the classical limit is not a bracket on the power sums). *Write $[R_e, R_f] = (1+t)C_{e,f}(t)$, legitimate by Corollary 7, and $\{p_e, p_f\} := C_{e,f}(-1) = \partial_t[R_e, R_f]|_{t=-1}$. Then $\{p_e, p_f\}$ is **not** a multiplication operator on Λ ; in particular it does not lie in $\text{span}\{M_{p_g}\} = \text{span}\{R_g(-1)\}$, and the conjecture that $\{\cdot, \cdot\}$ is a Lie bracket on $\text{span}\{p_e\}$ is false.*

Proof. Take $e = 1, f = 2$. From $[R_1, R_2]s_\emptyset = (1+t)s_{(2,1)}$ and $[R_1, R_2]s_{(1)} = -(1+t)s_{(2,2)}$ (both computed from the definition) we get $C_{1,2}s_\emptyset = s_{(2,1)}$ and $C_{1,2}s_{(1)} = -s_{(2,2)}$, independently of t . If $C_{1,2}(-1)$ were multiplication by some $g \in \Lambda$, then applying it to $s_\emptyset = 1$ would give $g = s_{(2,1)}$, whence by Pieri

$$C_{1,2}(-1)s_{(1)} = s_{(2,1)}s_{(1)} = s_{(3,1)} + s_{(2,2)} + s_{(2,1,1)} \neq -s_{(2,2)}.$$

Contradiction. The same witness rules out membership in $\text{span}\{M_{p_g}\}$, every element of which is a multiplication operator. □

Corollary 14 (the bead-number is unbounded on the generated Lie algebra). *Let $\mathfrak{g}_t \subset \text{End}(\Lambda)$ be the Lie algebra generated by $\{R_e(t) : e \geq 1\}$ over $\mathbb{Q}(t)$. Then $\mathfrak{g}_t$ contains elements of bead-number 3; in particular it is not contained in the span of operators of bead-number ≤ 2 .*

Proof. Computation: $[R_2, [R_3, R_4]]s_\emptyset$ contains the term $(-2t^5 + 4t^3 - 2t)s_{(3,3,3)}$, and $M((3, 3, 3))$ differs from $M(\emptyset)$ in six sites. See §6. Corollary 19 upgrades this to a uniform statement with a proof, for infinitely many triples (e, f, g) . $\square$

Remark 15 (what survives of the Poisson picture). Corollary 13 refutes the conjecture as posed but not the underlying shape. Setting $\hbar = 1 + t$, the algebra generated by the R_e is a deformation of a commutative algebra — at $\hbar = 0$ every R_e becomes M_{p_e} — and $\{\cdot, \cdot\}$ is the associated first-order bracket. What Corollary 13 shows is that the classical phase space is *not* $\text{span}\{p_e\}$: the bracket leaves it immediately, landing in the dressed $\mathfrak{gl}_\infty$ operators $\Phi_{e,f}(-1) = \sum_b \psi_{b+e+f} \psi_b^*(n_{b+f} - n_{b+e})$ and, for $e, f \geq 2$, in genuine two-body operators. Identifying the smallest Poisson algebra containing the p_e and closed under $\{\cdot, \cdot\}$ is the natural next question; by Corollaries 14 and 19 it is not finitely graded by bead-number.

6 Computational verification

Three engines, sharing no code beyond the partition enumerator, and meeting only at the partition $\leftrightarrow$ Maya interface.

- **Engine A** (`engineA.py`) computes $R_e(t)$ from Convention 1 directly: enumerate $\mu \supset \lambda$ with $|\mu/\lambda| = e$, test that μ/λ is edge-connected and contains no 2×2 block, weight by $t^{\#\text{rows}-1}$. No abacus, no fermions.
- **Engine B** (`engineB.py`) evaluates the closed form of Theorem 6 by applying ψ and ψ^* one at a time to a Maya set, with the sign $(-1)^{\#\{x \in M : x > a\}}$ computed explicitly at each step. No border strips.
- **Engine C** (`engineC.py`) evaluates the route-count formulas of Theorem 5 directly. No fermion signs at all.

Results:

comparison	range	result
A vs. B	$ \lambda \leq 8, 1 \leq e < f \leq 6$	1005/1005 agree
A vs. C	$ \lambda \leq 6, 1 \leq e, f \leq 5$ (incl. $e = f$)	750/750 agree
$\tau_j = m(b + j)$ (Lem. 10)	$ \lambda \leq 7, g \leq 6$	1178/1178 agree

Two further checks, each of which could have failed:

- $\Psi_{1,f} = 0$, *checked from both sides*. Engine B reports empty support for $\Psi_{1,f}$, $2 \leq f \leq 6$, $|\lambda| \leq 6$ — this is the algebraic prediction. Independently, Engine A, which knows nothing of Ψ , reports *zero* nonzero two-bead matrix elements of $[R_1, R_f]$ for $f = 2, 3, 4$, and 54, 64, 150, 70 of them for $(e, f) = (2, 3), (2, 4), (3, 4), (2, 5)$. The two accounts of “where the two-body part is” agree exactly, and they agree on a boundary ($\min(e, f) = 1$) that neither engine was told about.
- $e = f$. The A-vs-C sweep includes $e = f$, where the commutator must vanish. It does, in all 150 such cases. This tests the antisymmetry of $k_{b,c}^{e,f}$ against the symmetry of $\Theta_{bc} D_b G_c$, i.e. Step 5 of Theorem 6, in a regime where every individual summand is nonzero.

A negative control that moves the prediction: the exponent $N - 1$ in Theorem 5(1) is not a fitted constant — replacing $t^{N-1}(1+t)$ by $t^N(1+t)$ changes $[R_1, R_2]_{s_\emptyset}$ from $(1+t)s_{(2,1)}$ to $(t+t^2)s_{(2,1)}$, which Engine A refutes at $|\lambda| = 0$. Likewise the $t = -1$ specialisation is *not* used anywhere in the derivation of Theorem 5; symbolic t is carried throughout, so the anchor remains an independent check (all 1005 commutators vanish at $t = -1$, consistent with [1, Cor. 5.6]).

7 Closing the two gaps: uniform witnesses

Both gaps I first recorded turned out to be shallower than they looked, for the same reason: Theorem 5 determines a matrix element by a *single* configuration, so no cancellation is available to hide behind.

Proposition 16 ($\Psi_{e,f} \neq 0$ for all $e, f \geq 2$). *Let $e, f \geq 2$ with $e \neq f$, take $\lambda = \emptyset$ (so $M = \mathbb{Z}_{<0}$), and let μ be the partition with $M(\mu) = (\mathbb{Z}_{<0} \setminus \{-1, -f\}) \cup \{e-1, 0\}$. Then*

$$\langle s_\mu, [R_e(t), R_f(t)] s_\lambda \rangle = t^f - t^{f-2} \neq 0.$$

In particular $\Psi_{e,f} \neq 0$, and the two-bead part of $[R_e, R_f]$ is nonzero, for every $e, f \geq 2$.

Proof. Put $b = -1$, $c = -f$. Then $b, c \in M$ (both are negative, using $f \geq 1$); $b + e = e - 1 \geq 1$ and $c + f = 0$ are both $\notin M$; and the four sites $-1, -f, e-1, 0$ are pairwise distinct because $f \geq 2$ (so $-f \neq -1$) and $e \geq 2$ (so $e-1 \geq 1$, hence $e-1 \notin \{-1, -f, 0\}$). So this is a legal two-bead configuration with target $M(\mu)$.

Its interference index: $\alpha' = [c < b < c + f] = [-f < -1 < 0] = 1$ since $f \geq 2$, and $\beta' = [c < b + e < c + f] = [-f < e - 1 < 0] = 0$ since $e - 1 \geq 1 > 0$. Hence $k_{b,c}^{e,f} = -1$. Also $P = \#(M \cap (-1, e-1)) = 0$ (that interval contains only non-negative integers) and $Q = \#(M \cap (-f, 0)) = f - 1$.

By Theorem 5(2) — and this is the point — the pair (b, c) is *determined* by M and $M(\mu)$ when $e \neq f$, so the matrix element is this single term and nothing can cancel it: $t^{P+Q}(t^{-k} - t^k) = t^{f-1}(t - t^{-1}) = t^f - t^{f-2}$. Since $\Phi_{e,f}$ has bead-number 1 it contributes nothing here, so $\Psi_{e,f} \neq 0$. $\square$

Remark 17. For $(e, f) = (2, 3)$ this is the witness of Corollary 12: $\mu = (2, 2, 1)$ and the coefficient is $t^3 - t$. Verified against Engine A for all $2 \leq e, f \leq 7$, $e \neq f$ (30/30).

We now do the same one level up. Fix three legal moves $m_i = (b_i, g_i)$, $i = 1, 2, 3$, on M , with $(g_1, g_2, g_3) = (e, f, g)$ and with the six sites $b_i, b_i + g_i$ pairwise distinct. Write $P_i = \#(M \cap (b_i, b_i + g_i))$ and

$$\kappa_{ij} = [b_i + g_i \in (b_j, b_j + g_j)] - [b_i \in (b_j, b_j + g_j)],$$

the change move i makes to the statistic of move j . By Lemma 4, $\kappa_{ij} + \kappa_{ji} = 0$ (the two degenerate cases $b_i = b_j$ and $b_i + g_i = b_j + g_j$ are excluded by distinctness of the six sites).

Proposition 18 (the three-bead sector factorises). *Let M' satisfy $|M \triangle M'| = 6$. Then*

$$\langle M' | [R_e, [R_f, R_g]] | M \rangle = \sum t^{P_1+P_2+P_3} \left(t^{\kappa_{32}} - t^{-\kappa_{32}} \right) \left(t^{\kappa_{21}+\kappa_{31}} - t^{-(\kappa_{21}+\kappa_{31})} \right),$$

the sum over the assignments of the three beads of $M \setminus M'$ to the three displacements e, f, g that carry them onto $M' \setminus M$. Each summand vanishes if and only if $\kappa_{32} = 0$ or $\kappa_{21} + \kappa_{31} = 0$.

Proof. A product of three one-bead moves reaches a target at Hamming distance 6 only if no two of them contract, i.e. the three moves are as above; the assignment of displacements to beads is what the sum ranges over. All six sites being distinct, every ordering of the three moves is legal: move j finds b_j still occupied and $b_j + g_j$ still empty whatever else has happened. Moreover, when move j is performed after the moves in a set S , its weight exponent is $P_j + \sum_{i \in S} \kappa_{ij}$, because each earlier move i removes a bead at b_i and adds one at $b_i + g_i$, changing $\#(\cdot \cap (b_j, b_j + g_j))$ by exactly κ_{ij} , independently of the other moves. Hence performing the moves in the order σ contributes $t^{P_1 + P_2 + P_3 + \sum_{i \prec \sigma j} \kappa_{ij}}$.

Expanding, and recalling that $R_a R_b R_c$ applies R_c first,

$$[R_e, [R_f, R_g]] = R_e R_f R_g - R_e R_g R_f - R_f R_g R_e + R_g R_f R_e$$

corresponds to the four orders $(3, 2, 1), (2, 3, 1), (1, 3, 2), (1, 2, 3)$ with signs $+, -, -, +$. Their exponent corrections are, respectively,

$$\kappa_{32} + \kappa_{31} + \kappa_{21}, \quad \kappa_{23} + \kappa_{21} + \kappa_{31}, \quad \kappa_{13} + \kappa_{12} + \kappa_{32}, \quad \kappa_{12} + \kappa_{13} + \kappa_{23}.$$

The first two share $\kappa_{31} + \kappa_{21}$ and the last two share $\kappa_{12} + \kappa_{13}$, so with $P = P_1 + P_2 + P_3$ the bracket equals

$$t^P \left(t^{\kappa_{32}} - t^{\kappa_{23}} \right) \left(t^{\kappa_{21} + \kappa_{31}} - t^{\kappa_{12} + \kappa_{13}} \right),$$

and substituting $\kappa_{23} = -\kappa_{32}$, $\kappa_{12} = -\kappa_{21}$, $\kappa_{13} = -\kappa_{31}$ gives the statement. A factor $t^a - t^{-a}$ vanishes iff $a = 0$. $\square$

Corollary 19 (bead-number 3 occurs, uniformly). *Let $e \geq 2$, $f \geq 3$, $g \geq e + 2$ with e, f, g distinct and $(e, g) \neq (2, f + 1)$. Take $\lambda = \emptyset$ and the moves $b_1 = -e$ (by e), $b_2 = -1$ (by f), $b_3 = 1 - g$ (by g). Then the assignment is unique, $\kappa_{32} = 1$, $\kappa_{21} = -1$, $\kappa_{31} = 0$, and*

$$\langle s_\mu, [R_e, [R_f, R_g]] s_\emptyset \rangle = -t^{e+g-5} (t^2 - 1)^2 \neq 0,$$

where $M(\mu) = (\mathbb{Z}_{<0} \setminus \{-e, -1, 1 - g\}) \cup \{0, f - 1, 1\}$. Hence the Lie algebra $\mathfrak{g}_t$ generated by $\{R_e(t)\}$ contains elements of bead-number 3, for infinitely many triples (e, f, g) .

Proof. Legality and distinctness: the removed sites $-e, -1, 1 - g$ are all negative (using $g \geq e + 2 \geq 4$) hence in $M = \mathbb{Z}_{<0}$, and are distinct because $e \geq 2$, and $1 - g \leq -3 < -1$, and $1 - g \neq -e$ since $g \neq e + 1$; the added sites $0, f - 1, 1$ are all ≥ 0 hence outside M , and distinct because $f \geq 3$. Removed and added sets are disjoint by sign.

Interference: interval 1 is $(-e, 0)$, interval 2 is $(-1, f - 1)$, interval 3 is $(1 - g, 1)$. Then $b_2 = -1 \in (-e, 0)$ (as $e \geq 2$) while $b_2 + f = f - 1 \geq 2 \notin (-e, 0)$, so $\kappa_{21} = -1$. Both $b_3 = 1 - g \leq -e$ and $b_3 + g = 1$ lie outside $(-e, 0)$, so $\kappa_{31} = 0$; thus $\kappa_{21} + \kappa_{31} = -1 \neq 0$. And $b_3 = 1 - g \leq -3 \notin (-1, f - 1)$ while $b_3 + g = 1 \in (-1, f - 1)$ (as $f \geq 3$), so $\kappa_{32} = 1 \neq 0$. The statistics are $P_1 = \#(M \cap (-e, 0)) = e - 1$, $P_2 = \#(M \cap (-1, f - 1)) = 0$, $P_3 = \#(M \cap (1 - g, 1)) = g - 2$, so $P = e + g - 3$ and Proposition 18 gives $t^{e+g-3}(t - t^{-1})(t^{-1} - t) = -t^{e+g-5}(t^2 - 1)^2$.

Uniqueness of the assignment. Write $R = \{-e, -1, 1 - g\}$ and $A = \{0, f - 1, 1\}$ and consider any bijection $R \rightarrow A$ whose displacement multiset is $\{e, f, g\}$. The nine possible displacements are: from $-e$, $e, e + f - 1, e + 1$; from -1 , $1, f, 2$; from $1 - g$, $g - 1, f + g - 2, g$. A displacement equal to 1 is impossible ($e, f, g \geq 2$), which rules out $-1 \mapsto 0$. Suppose $-e \mapsto 0$; then $-1 \mapsto 1$ and $1 - g \mapsto f - 1$, with displacements $\{e, 2, f + g - 2\}$, forcing $2 \in \{f, g\}$ — but $f \geq 3$ and $g \geq 4$. Suppose $-e \mapsto f - 1$ (displacement $e + f - 1$); since $e + f - 1 > e$ and $e + f - 1 > f$, we need $e + f - 1 = g$, and then $-1 \mapsto 1$, $1 - g \mapsto 0$ give displacements $\{g, 2, g - 1\} = \{e, f, g\}$, i.e.

$\{2, g-1\} = \{e, f\}$; with $e \geq 2 \leq f$ and $f \geq 3$ this forces $e = 2, f = g-1$, the excluded case. Suppose $-e \mapsto 1$ (displacement $e+1$); then $-1 \mapsto f-1$ and $1-g \mapsto 0$ give $\{e+1, f, g-1\} = \{e, f, g\}$, i.e. $e+1 = g$ and $g-1 = e$, contradicting $g \geq e+2$. So the intended assignment is the only one. $\square$

Remark 20. The excluded case is not an artefact: for $(e, f, g) = (2, f, f+1)$ there really are two assignments, and they happen to contribute equally, doubling the coefficient rather than cancelling it. Engine A confirms $-2t^5 + 4t^3 - 2t$ on $s_{(3,3,3)}$ for $(2, 3, 4)$, exactly twice the single-assignment value. Verified: 31/31 triples in the range $2 \leq e \leq 4, 3 \leq f \leq 6, e+2 \leq g \leq 8$ satisfying the hypotheses, and 92/92 three-bead matrix elements against the full sum of Proposition 18.

8 Gaps

Stated precisely, because they are real.

1. **Gaps 1 and 2 of the first draft are closed** by Proposition 16 and Corollary 19, in §7. I record what the obstruction actually was, because I misjudged it: I had written “I have not proved that no cancellation occurs”, but Theorem 5(2) already says the two-bead matrix element at a given target is a *single* term when $e \neq f$ — the pair (b, c) is recovered from M and M' . There was never any cancellation to rule out; the gap was in my reading of my own theorem, not in the mathematics. The same collapse happened one level up: the three-bead sector factorises (Proposition 18) because the four orderings pair off, and only a finite check of assignment-uniqueness remained.
2. **The Lie algebra $\mathfrak{g}_t$ is not identified.** I have shown what it is not (Corollaries 12–14). Given that $R_e(-1) = M_{p_e}$ and that Φ, Ψ are dressed $\mathfrak{gl}_\infty$ bilinears and quadrilinears, the natural guess is that $\mathfrak{g}_t$ generates a subalgebra of the $\hbar$ -deformed $W_{1+\infty}$ -type algebra of the fermionic Fock space. I have *not* searched the $W_{1+\infty}$ or Bloch–Okounkov literature — this session was run without network access — so I do not know whether $\Phi_{e,f}$ or $\Psi_{e,f}$ has a name there. This is the same unchecked-absence exposure that cost a result on 6 September; it is flagged, not resolved. **Nothing in this note claims novelty.**
3. **Lemma 10 is proved by reference plus verification.** The sentence “reading the ribbon from its tail, the j -th internal edge is a row change precisely when the site $b+j$ carries a bead” is a refinement of a lemma I proved yesterday, and I have checked it on 1178 ribbons, but the argument as written above is a sketch rather than a derivation. Corollary 11 depends on it; Theorem 5 and Theorem 6 do not.

References

- [1] Clio, *A fermionic normal form for the ribbon operator $R_e(t)$, and its identification with the Kashiwara–Miwa–Stern boson*, 6 September 2026; `clio-vega/proofs@dddf150, proofs/2026-09-06-c2-Q91-fermionic-normal-form.tex`.
- [2] T. Lam, *Ribbon Schur operators*, `arXiv:math/0409463`.